

Smart Public Transport Analytics Project Documentation

Teyzix Core - Data Analytics Internship - Task DA-3

1. Dataset Creation Method

The dataset used in this project was created originally for the task; no public or third-party dataset was used. It was generated programmatically in Python to realistically simulate the daily operations of a public transport network, ensuring the data was both original and representative of genuine operational patterns.

The generation process modelled real-world relationships rather than producing purely random values:

- Eight routes were defined, each with a realistic fare level (e.g. the airport route carries a premium fare).
- Departure times were weighted toward morning and evening commuter peaks, so ridership reflects real demand patterns.
- Delays were modelled to increase during poor weather and at peak hours, reproducing genuine congestion effects.
- Passenger counts were bounded by each bus type's seating capacity, and revenue was computed as passengers multiplied by the route fare.
- Fuel consumption reflected trip length and bus type, with Electric buses using near-zero fuel.

A total of 1,300 base records were generated, spanning four months (January–April 2025) across eight routes and forty drivers. Deliberate data-quality issues were then injected to create a realistic raw dataset requiring cleaning.

2. Data Dictionary

The dataset contains 15 fields describing each trip:

Field	Type	Description	Example
Route ID	Text	Unique identifier for each route (R01–R08)	R08
Route Name	Text	Descriptive name of the route	South Terminal – Business District
Bus Number	Text	Identifier of the specific bus vehicle	BUS-309
Bus Type	Text	Category of bus (Standard, AC, Electric, Mini)	Standard
Date	Date	Date of the trip (YYYY-MM-DD)	2025-03-16
Departure Time	Time	Scheduled departure time (24-hour)	16:30
Arrival Time	Time	Actual arrival time (24-hour)	17:13
Delay Duration	Integer	Delay in minutes beyond schedule	5
Passenger Count	Integer	Number of passengers on the trip	33
Ticket Revenue	Decimal	Total fare revenue for the trip (Rs)	2200.00
Fuel Consumption	Decimal	Fuel used for the trip (litres)	6.92
Driver ID	Text	Identifier of the assigned driver (D001–D040)	D005
Weather Condition	Text	Weather during the trip	Clear
Day Type	Text	Weekday or Weekend	Weekend
Trip Status	Text	Outcome of the trip (Completed, Delayed, Cancelled)	Completed

3. Data Cleaning Process

The raw dataset contained six categories of data-quality issues, each addressed systematically in Python (Pandas):

1. **Duplicate removal** — 35 fully duplicated rows were identified and removed.
2. **Missing value handling** — numeric fields (passengers, delay) filled with the median; fuel filled with the per-bus-type median; weather filled with the mode; missing driver IDs set to “Unknown”.
3. **Standardization** — 15 bus-type variants (e.g. “A/C”, “a.c.”, “Air Conditioned”) and 13 weather variants were mapped to clean categories; time and date formats were unified.
4. **Data type correction** — text-encoded numbers such as “Rs 2,200” and “6.9 L” were stripped of symbols and converted to numeric types; dates parsed to a single date type.
5. **Data validation** — trips with arrival before departure were corrected by swapping the times; negative delays were treated as missing.
6. **Outlier detection** — passenger counts above bus capacity were capped; negative counts and absurd delays removed; extreme fuel values flagged using a 3× IQR threshold.

The cleaned dataset contains 1,300 validated records with zero missing values, zero duplicates, standardised categories, and correct data types.

4. Dashboard Development

An interactive dashboard was developed in Microsoft Power BI. The cleaned CSV was imported, and calculated columns (Hour, Capacity) and DAX measures (for each KPI) were created to support the visuals.

- Executive summary: eight KPI cards across the top of the dashboard.
- Seven distinct chart types were used — horizontal bar (route performance), line (passenger trends), area (revenue), column (delay by weather), histogram-style column (peak hours), donut (fuel by bus type), and scatter (route comparison).
- Four slicers (Route, Trip Status, Weather, Bus Type) provide interactivity, filtering every visual simultaneously.

The dashboard was exported as both a Power BI file and a PDF, with screenshots captured of the interactive version.

5. KPI Definitions

Nine key performance indicators were developed to measure the health and efficiency of the transport operation. Each KPI was implemented as a DAX measure in Power BI and precisely defined to ensure consistent, unambiguous interpretation. The table below sets out the exact calculation behind each metric.

KPI	Definition
Total Passengers	Sum of Passenger Count across all trips.
Total Revenue	Sum of Ticket Revenue across all trips.
Average Daily Passengers	Total passengers divided by the number of distinct operating days.
Average Delay Time	Mean of Delay Duration across all trips (minutes).
Average Occupancy Rate	Mean of (Passenger Count ÷ bus capacity) across trips.
Trip Completion Rate	Share of trips not cancelled (Completed + Delayed) ÷ all trips.
Most Profitable Route	Route with the highest total Ticket Revenue.
Least Performing Route	Route with the lowest total Ticket Revenue.
Peak Travel Hour	Departure hour with the highest total passengers.

Note on Trip Completion Rate: this metric can be defined in two ways — strictly (only trips marked “Completed”) or operationally (all trips except those cancelled). The operational definition was adopted here, since a delayed trip still runs and serves its passengers; only a cancelled trip represents a genuine service failure. Under this definition the completion rate is 91.7%, versus 70.7% under the stricter definition.

6. Challenges Faced

- **Generating internally consistent data.** Ensuring arrival times, delays, revenue, and capacity all reconciled with one another required careful modelling rather than independent random values.
- **Cleaning diverse data-quality issues.** The raw data mixed missing values, duplicates, inconsistent labels, mixed formats, and impossible values — each needing a different, well-reasoned treatment.
- **Defining ambiguous KPIs.** “Trip completion rate” could be defined strictly (completed only) or operationally (not cancelled); the operational definition was chosen and documented.
- **Handling time formats in Power BI.** Extracting the hour from text-based time fields required a calculated column using TIMEVALUE.

7. Future Improvements

- **Geographic mapping.** Adding route coordinates would enable map-based visuals of ridership and delays across the city.
- **Predictive analytics.** A forecasting model could predict passenger demand or delays from weather and time, supporting proactive planning.
- **Real-time data integration.** Connecting to live operational feeds would turn the dashboard into a real-time monitoring tool.
- **Driver and vehicle analytics.** Deeper analysis of individual driver and vehicle performance could surface further efficiency opportunities.